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🌳 Tree BFS (Level Order Traversal) — Complete Interview Handbook

Pattern #14 of 28 | DSA Pattern Mastery Series Audience: Senior/Staff SWE interviews (Google, Amazon, Meta, Microsoft, Uber, Airbnb, Atlassian, Grab, Careem, Noon, Talabat, Dubai/Saudi & India Tier-1/Tier-2 product companies) **Companion:** Pairs with Striver's A2Z DSA Course — Binary Tree section

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SECTION 1 — Pattern Overview

1.1 What Is This Pattern?

Tree BFS (Breadth-First Search / Level Order Traversal) visits a tree **level by level**, using a **queue** to process all nodes at depth `d` before any node at depth `d+1`. This is the natural traversal for any problem that cares about "level," "layer," or "distance from root" rather than depth-first branch exploration.

1.2 Why Was This Pattern Invented?

Some tree problems fundamentally require grouping or comparing nodes **by their depth** (e.g., "print each level," "find the average value per level," "find the rightmost node visible per level"). DFS doesn't naturally group by level — a queue-based, level-synchronized traversal is needed, processing nodes in the exact order they're "discovered" by distance from the root, which is precisely what a FIFO queue provides.

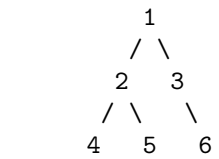
1.3 Real Intuition Behind The Pattern

Imagine a **rumor spreading through a social network, one "hop" at a time** — first the person who started it (level 0), then everyone they directly told (level 1), then everyone those people told next (level 2), and so on. A queue naturally enforces this "process everyone at the current hop before moving to the next hop" behavior.

1.4 Mental Model

A queue holds "the frontier" — all nodes discovered but not yet processed, always ordered such that shallower nodes are dequeued before deeper ones. Snapshotting the queue's size at the start of each level lets you process "exactly this level's nodes" as a distinct batch before enqueueing the next level.

1.5 Visual Explanation




```
BFS Queue evolution:
Start: queue=[1]
Level 0: dequeue 1, process → enqueue 2,3 → queue=[2,3]
Level 1: dequeue 2, process; dequeue 3, process → enqueue 4,5,6 → queue=[4,5,6]
Level 2: dequeue 4,5,6, process (no children) → queue=[]

Output by level: [[1], [2,3], [4,5,6]]
```

1.6 Simple Analogy

Tree BFS is like an **assembly line inspector who checks every item currently on the belt before letting the next batch move onto the belt** — never letting a "later" batch (deeper level) get ahead of an "earlier" one (shallower level).

1.7 When Should I Immediately Think About Using This Pattern?

- "Level order traversal," "print each level."

- “Average/sum/max per level.”
- “Rightmost/leftmost node per level” (right/left side view).
- “Minimum depth” (BFS finds the *shortest* path to a leaf faster than DFS in many cases, since it terminates the moment the first leaf is found at the shallowest depth).
- “Connect next pointers” between nodes on the same level.

SECTION 2 — Recognition Guide

2.1 Keywords

Keyword	Signal
“level order”	Direct signal
“level by level,” “layer by layer”	Direct signal
“average/sum per level”	Level-batched BFS
“right side view,” “left side view”	BFS tracking last/first node per level
“minimum depth”	BFS for shortest-path-to-leaf (early termination advantage)
“connect same-level nodes” (next pointers)	BFS-based level linking

2.2 Hidden Hints

Any mention of “level” or “depth” as a grouping criterion (not just a single aggregate over the whole tree) is the strongest tell for BFS over DFS.

2.3 Interview Clues

Interviewer asks for output structured “as a list of lists, one per level,” or asks specifically about the shortest path/minimum depth (where BFS’s level-synchronized nature gives an efficient early-exit advantage over DFS).

2.4 Common Trick Words

“Zigzag” (level order with alternating direction), “vertical order” (BFS combined with coordinate tracking), “connect” (populating next-right pointers per level).

2.5 What Interviewers Expect

Correct use of the **queue-size-snapshot** technique to process exactly one level per iteration (a common point of confusion for candidates new to this pattern), and recognition of when BFS’s early-termination advantage (for shortest-path/minimum-depth problems) beats DFS.

2.6 When NOT To Use This Pattern

- The problem needs **bottom-up aggregation** (height, diameter, sum) with no level-grouping requirement — DFS is more natural and often more memory-efficient (O(h) vs O(w) space).
- The tree is very **wide** (many nodes per level) — BFS’s O(w) space (w = max width) could exceed DFS’s O(h) space (h = height) for wide, shallow trees; consider this trade-off explicitly.
- You need **path-based** information (root-to-node path, path sum) — DFS naturally carries path state down the recursion; BFS would need extra bookkeeping (e.g., storing parent pointers) to reconstruct paths.

SECTION 3 — Decision Framework

```
Does the problem require LEVEL-BY-LEVEL grouping or per-level aggregation?

       Yes → USE BFS (queue-based level order traversal)

        No

Does it need the SHORTEST PATH / MINIMUM DEPTH with early-termination benefit?

       Yes → USE BFS (terminates as soon as the shallowest valid node/leaf is found)

        No

Does it need BOTTOM-UP AGGREGATION (height, sum, diameter) or PATH TRACKING (root-to-node)?

       Yes → USE DFS (Pattern #13) instead - more natural, often less space for narrow/deep trees

        No

Is the tree very WIDE (many nodes per level) vs. very DEEP?

      Wide → DFS's O(h) space may be preferable to BFS's O(w) space
      Deep → BFS's O(w) space may be preferable to DFS's O(h) space (avoids deep recursion)
```

Why: BFS and DFS both visit every node in O(n) time, but they differ fundamentally in **what grouping/order they naturally provide** (level-batches vs. depth-first branches) and in **their space complexity's dependency on tree shape** (width vs. height) — choosing based on the problem's actual structural need, not habit, is the key decision.

SECTION 4 — Why This Pattern Works

Mathematical/Logical: A queue is a FIFO (first-in-first-out) structure. If nodes are enqueued in strictly non-decreasing order of depth (which BFS guarantees, since a node's children are always enqueued immediately after the node itself is dequeued, and all same-depth nodes are enqueued before any deeper node), then dequeuing always processes nodes in non-decreasing depth order — this FIFO property is exactly what produces the level-by-level guarantee.

Intuitive: Because you only ever add a node's children right after processing that node, and you process nodes in the order they were added, no node from a deeper level can "jump ahead" of a node from a shallower level — the queue naturally acts as a "wave front" expanding outward one level at a time.

Correctness Proof: *Invariant:* at the start of processing level `d`, the queue contains exactly the nodes at depth `d`, and no nodes from depth `> d` have been enqueued yet. *Base case:* initially, the queue contains just the root (depth 0) — trivially true. *Inductive step:* processing all nodes currently in the queue (a snapshotted count, all at depth `d`) and enqueueing their children (all at depth `d+1`) transitions the queue to containing exactly depth `d+1` nodes, preserving the invariant. *Termination:* when the queue becomes empty, all nodes have been visited, level by level. **QED.**

SECTION 5 — Algorithm Framework

5.1 Step-by-Step Framework

1. Initialize a queue with the root node (if non-null).

2. While the queue is non-empty: **snapshot the current queue size** (this is exactly the number of nodes at the current level).
3. Dequeue exactly that many nodes, processing each and enqueueing their non-null children.
4. Repeat until the queue is empty.

5.2 General Template

```
function levelOrder(root):
    if root is null: return []
    result = []
    queue = [root]

    while queue is not empty:
        levelSize = length(queue)
        currentLevel = []

        for i in range(0, levelSize):
            node = queue.dequeue()
            currentLevel.append(node.value)
            if node.left is not null: queue.enqueue(node.left)
            if node.right is not null: queue.enqueue(node.right)

        result.append(currentLevel)

    return result
```

5.3 Minimum Depth Template (Early Termination)

```
function minDepth(root):
    if root is null: return 0
    queue = [(root, 1)]
    while queue is not empty:
        node, depth = queue.dequeue()
        if node.left is null and node.right is null:
            return depth                              # first leaf found = minimum depth
        if node.left is not null: queue.enqueue((node.left, depth + 1))
        if node.right is not null: queue.enqueue((node.right, depth + 1))
```

5.4 Interview Thinking Process

1. "This needs level-by-level grouping (or shortest-path early termination) — I'll use BFS with a queue, not DFS."
2. "I'll snapshot the queue's size at the start of each iteration to know exactly how many nodes belong to the current level."
3. "I'll enqueue children only after fully processing the current level's snapshot, keeping levels cleanly separated."
4. "For minimum depth, I'll return immediately upon finding the first leaf — BFS's level-synchronized nature guarantees this is the shallowest one."

SECTION 6 — Time & Space Complexity

Case	Time	Space	Why
Worst Case	O(n)	O(w), w = maximum width (nodes at the widest level)	Every node visited once; queue holds at most one full level at a time
Average Case	O(n)	O(n/2) for a complete binary tree's last level (still O(n) in the worst case for width)	Balanced trees can have widths approaching n/2 at the last level
Best Case	O(1) to O(n) for early-termination problems (minimum depth)	O(w)	Early exit possible the moment the shallowest qualifying node is found
Amortized	O(n) total across the single traversal	O(w)	No repeated work — each node enqueued/dequeued exactly once

Width vs. height trade-off: for a **complete binary tree**, the last level can contain up to `n/2` nodes — BFS space can approach O(n) in the worst case, potentially worse than DFS's O(log n) for a balanced tree; always mention this trade-off explicitly.

SECTION 7 — Edge Cases

Edge Case	Example	Handling
Empty tree	`root = null`	Return empty result immediately
Single node	`[5]`	One level, containing just the root
Completely skewed tree (linked-list-like)	Every node has only one child	Each "level" contains exactly one node — BFS space stays O(1) here, better than the intuition of O(w) suggests for this specific shape
Very wide tree (complete binary tree)	Last level has ~n/2 nodes	BFS space approaches O(n) — worth mentioning as a trade-off vs DFS
Need zigzag/alternating direction output	"Zigzag Level Order Traversal"	Reverse every other level's collected list before appending to the result
Need only the last level	"Find Bottom Left Tree Value"	Track the last-processed level, or the first node encountered per level using right-to-left traversal order
n-ary tree (more than 2 children)	Tree with variable children count	Enqueue all children (not just left/right) — the core algorithm generalizes directly

Common mistakes: forgetting to snapshot the queue size before the inner loop (causing children enqueued during processing to be incorrectly included in the *current* level's count, since the

queue's size changes dynamically as you enqueue); forgetting to check for null children before enqueueing, causing null-pointer errors downstream.

SECTION 8 — Pros & Cons

Advantages: Naturally produces level-grouped output; provides an efficient early-termination advantage for shortest-path/minimum-depth problems; simple and predictable O(n) time. **Disadvantages:** O(w) space can be worse than DFS's O(h) space for wide, shallow trees (e.g., complete binary trees where the last level holds ~n/2 nodes); doesn't naturally carry "path so far" information without extra bookkeeping. **Trade-offs:** BFS (O(w) space, natural level-grouping, early-exit shortest path) vs. DFS (O(h) space, natural bottom-up aggregation, natural path tracking) — choose based on which structural need (levels vs. depth-aggregation/paths) the problem actually has. **Limitations:** Reconstructing root-to-node paths requires additional parent-pointer tracking, which DFS gets "for free" via the call stack. **Inefficient when:** the tree is very wide (space cost approaches O(n)) and only depth-aggregation (not level-grouping) is actually needed — DFS would be more space-efficient in that specific case.

SECTION 9 — Real World Applications

Domain	Application
Networking	Broadcast/flooding protocols in network topology discovery process nodes hop-by-hop, exactly like BFS levels
Social Networks (Meta, LinkedIn)	"Degrees of separation" / mutual friend suggestions computed via BFS from a user node
Google/Search	Web crawlers often use BFS-like strategies to explore links level-by-level from seed URLs
Organizational Systems	Reporting-hierarchy analysis by management level (e.g., "list all employees at each reporting depth")
Game Development	Fog-of-war/visibility expansion in strategy games often uses BFS from the player's position, hop by hop
GPS/Navigation	Unweighted shortest-path-hop-count estimation (before applying weighted shortest path algorithms)
Distributed Systems	Gossip protocols propagate information level-by-level through a cluster, analogous to BFS waves
UI/UX Rendering	Rendering nested component trees level-by-level for certain layout/paint algorithms
Database Systems	B-Tree level-order traversal for bulk-loading or level-based index analysis

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SECTION 10 — Interview Strategy

How seniors answer: They immediately state whether the problem needs level-grouping or depth-aggregation, correctly explain the queue-size-snapshot technique to cleanly separate levels, and proactively mention the width-vs-height space trade-off between BFS and DFS.

How juniors answer: They often forget to snapshot the queue size, leading to levels bleeding into each other, or they default to BFS out of habit even when the problem is really asking for a DFS-style bottom-up aggregation (unnecessarily complicating the solution).

Typical follow-ups: "Can you do a zigzag traversal?" "How would you find the rightmost node at each level?" "What's the space complexity, precisely, in terms of tree width?" "Would DFS be more space-efficient here — why or why not?"

Optimization questions: "Can you reduce space usage for a very wide tree?" (Generally the O(w) cost is inherent to level-order traversal; discuss whether the problem can be reframed to avoid needing full-level grouping.)

SECTION 11 — Pattern Variations

Variation	Description	Example
Basic Level Order	Collect nodes level by level	Binary Tree Level Order Traversal
Bottom-Up Level Order	Levels collected bottom-to-top	Binary Tree Level Order Traversal II
Zigzag Level Order	Alternate direction per level	Binary Tree Zigzag Level Order Traversal
Right/Left Side View	Track last/first node per level	Binary Tree Right Side View
Average/Sum Per Level	Aggregate values within each level	Average of Levels in Binary Tree
Minimum Depth (Early Exit)	BFS terminates at first leaf found	Minimum Depth of Binary Tree
Connect Next Pointers	Link nodes within the same level	Populating Next Right Pointers in Each Node
N-ary Tree BFS	Generalizes to more than 2 children per node	N-ary Tree Level Order Traversal

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SECTION 12 — Comparison With Other Patterns

Pattern	Difference	When To Prefer
Tree DFS	Depth-first, natural for bottom-up aggregation and path tracking	Height/diameter/sum computations, root-to-node paths
Graph BFS	Generalizes to structures with cycles, requiring visited-tracking	Data isn't strictly a tree (has cycles)
Dijkstra's Algorithm	BFS generalized to weighted edges via a priority queue	Edge weights aren't uniform (BFS assumes unweighted/unit-cost edges)

Comparison Table

Aspect	Tree BFS	Tree DFS
Traversal order	Level-by-level	Depth-first (branch by branch)
Space	O(w) — max width	O(h) — height
Natural for	Level grouping, shortest path (unweighted)	Bottom-up aggregation, path tracking
Early-exit advantage	Yes, for shortest-path/min-depth problems	No (must potentially explore a full branch before backtracking)

SECTION 13 — Problem Classification

Difficulty	Characteristics	Examples
Easy	Basic level order collection	Binary Tree Level Order Traversal, Average of Levels in Binary Tree
Medium	Directional variants, side views	Zigzag Level Order Traversal, Binary Tree Right Side View, Populating Next Right Pointers
Hard	Combined with coordinate tracking or complex aggregation	Vertical Order Traversal of a Binary Tree, Cousins in Binary Tree II
Very Hard	Multi-tree BFS combinations, advanced state per level	Binary Tree Maximum Width with overflow-safe indexing, complex n-ary tree BFS with multi-attribute aggregation

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SECTION 14 — 30 Interview Problems

#	Problem	Difficulty	Companies	Why Pattern Applies	Learning Objective
1	Binary Tree Level Order Traversal	Medium	Amazon, Meta, Microsoft, Google	Direct level-batched BFS	Foundational mechanics
2	Binary Tree Level Order Traversal II	Medium	Amazon, Meta	Bottom-up level collection	Reverse-order variant
3	Binary Tree Zigzag Level Order Traversal	Medium	Amazon, Meta, Microsoft	Alternating direction per level	Directional variant

#	Problem	Difficulty	Companies	Why Pattern Applies	Learning Objective
4	Average of Levels in Binary Tree	Easy	Amazon, Google	Per-level aggregation	Aggregation variant
5	Binary Tree Right Side View	Medium	Amazon, Meta, Google, Microsoft	Last node per level tracking	Side-view variant
6	Binary Tree Left Side View (variant)	Medium	Amazon, Meta	First node per level tracking	Side-view variant
7	Minimum Depth of Binary Tree	Easy	Amazon, Meta	Early-termination BFS	Shortest-path BFS advantage
8	Populating Next Right Pointers in Each Node	Medium	Amazon, Meta, Microsoft	Level-based pointer linking	Level-based construction
9	Populating Next Right Pointers in Each Node II	Medium	Amazon, Meta	Level-based pointer linking (imperfect tree)	Advanced level linking
10	N-ary Tree Level Order Traversal	Medium	Amazon, Google	BFS generalized to multiple children	N-ary generalization
11	Maximum Width of Binary Tree	Medium	Amazon, Meta, Google	BFS with coordi-nate/index tracking	Coordinate-augmented BFS
12	Vertical Order Traversal of a Binary Tree	Hard	Amazon, Meta, Google	BFS/DFS with coordinate tracking + sorting	Coordinate-augmented traversal
13	Cousins in Binary Tree	Easy	Amazon, Meta	Level + parent tracking via BFS	Level + relationship tracking
14	Cousins in Binary Tree II	Medium	Amazon, Google	Level-based sum aggregation with exclusions	Advanced level aggregation
15	Find Bottom Left Tree Value	Medium	Amazon, Google	Level order with first-node tracking	Directional level tracking

#	Problem	Difficulty	Companies	Why Pattern Applies	Learning Objective
16	Binary Tree Level Order Traversal (with N-ary variant)	Medium	Google	Multi-child level order	N-ary BFS reinforcement
17	Deepest Leaves Sum	Medium	Amazon, Google	Level-based aggregation of last level	Last-level aggregation
18	All Nodes Distance K in Binary Tree	Medium	Amazon, Meta, Google	BFS from a target node after building parent pointers	Cross-pattern (DFS + BFS combination)
19	Rotting Oranges (contrast, grid BFS)	Medium	Amazon, Google	Contrast: Graph/Grid BFS, not tree BFS	Pattern-boundary awareness
20	Binary Tree Paths (contrast)	Easy	Amazon	Contrast: DFS is more natural here	Pattern-boundary awareness
21	Symmetric Tree (BFS variant)	Easy	Amazon, Meta	BFS-based mirror comparison alternative to DFS	Alternative traversal approach
22	Check Completeness of a Binary Tree	Medium	Amazon, Google, Meta	BFS with null-marker detection	Structural validation via BFS
23	Serialize and Deserialize N-ary Tree	Hard	Amazon, Google	BFS-based serialization alternative	Constructive BFS
24	Binary Tree Vertical Order Traversal (contrast with Zigzag)	Hard	Amazon, Meta	Coordinate + BFS combination	Advanced coordinate BFS
25	Add One Row to Tree	Medium	Amazon	Level-targeted BFS modification	Level-targeted mutation
26	Even Odd Tree	Medium	Amazon, Google	Level-based validity checking (parity + monotonicity)	Level-based validation
27	Sum of Nodes with Even-Valued Grandparent (contrast, DFS preferred)	Medium	Amazon	Contrast: DFS more natural for ancestor relationships	Pattern-boundary awareness

#	Problem	Difficulty	Companies	Why Pattern Applies	Learning Objective
28	Binary Tree Coloring Game (contrast, DFS-based)	Medium	Google	Contrast: DFS for subtree size computation	Pattern-boundary awareness
29	Step-By-Step Directions From a Binary Tree Node to Another	Medium	Amazon, Google	BFS/DFS hybrid for path finding via LCA	Cross-pattern combination
30	Reorder Routes to Make All Paths Lead to the City Zero (contrast, graph BFS/DFS)	Medium	Amazon, Google	Contrast: Graph BFS/DFS, not tree BFS	Pattern-boundary awareness

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SECTION 15 — Common Mistakes

1. Forgetting to snapshot the queue size before the inner processing loop, causing children enqueued mid-level to be incorrectly counted as part of the current level. *Fix:* always capture `levelSize = queue.length` before the inner loop begins.
2. Forgetting to check for null children before enqueueing, causing null-pointer errors or incorrect level counts. *Fix:* always guard `if node.left is not null` / `if node.right is not null`.
3. Using BFS out of habit for problems that are actually more naturally and efficiently solved with DFS (bottom-up aggregation, path tracking), adding unnecessary complexity. *Fix:* always ask "does this need level-grouping, or depth-aggregation/paths?" before choosing.
4. Not considering the space trade-off (O(w) vs O(h)) explicitly when discussing complexity — missing an opportunity to demonstrate deeper understanding. *Fix:* proactively mention this trade-off.
5. In zigzag/directional variants, reversing the wrong levels or reversing at the wrong stage (before vs. after collecting the level's values). *Fix:* collect the level normally first, then conditionally reverse based on the level's parity.

Why people fail: the queue-size-snapshot technique is the one non-obvious "trick" in an otherwise simple algorithm, and candidates who don't understand *why* it's needed (the queue's size changes dynamically as you enqueue children) often get subtly wrong level groupings that still produce a flattened, seemingly-plausible-looking output.

SECTION 16 — Optimization Techniques

- **Time:** Already optimal at O(n) — focus on avoiding redundant extra passes (e.g., don't compute tree height separately if it can be derived from the number of BFS levels processed).
- **Space:** For extremely wide trees, consider whether the problem can be reframed to avoid full-level materialization (e.g., only tracking the last node per level, not the entire level's contents, for "right side view").

- **Readability:** Clearly comment the queue-size-snapshot step's purpose; use descriptive variable names (`levelSize`, `currentLevel`) rather than generic `n`, `temp`.
- **Interview performance:** Explicitly state the width-vs-height space trade-off between BFS and DFS when discussing complexity — this small addition signals deeper structural understanding.

SECTION 17 — Multiple Language Templates

Java

```java
public List<List<Integer>> levelOrder(TreeNode root) {
    List<List<Integer>> result = new ArrayList<>();
    if (root == null) return result;
    Queue<TreeNode> queue = new LinkedList<>();
    queue.offer(root);
    while (!queue.isEmpty()) {
        int levelSize = queue.size();
        List<Integer> currentLevel = new ArrayList<>();
        for (int i = 0; i < levelSize; i++) {
            TreeNode node = queue.poll();
            currentLevel.add(node.val);
            if (node.left != null) queue.offer(node.left);
            if (node.right != null) queue.offer(node.right);
        }
        result.add(currentLevel);
    }
    return result;
}
```

JavaScript

```javascript
function levelOrder(root) {
    if (!root) return [];
    const result = [];
    const queue = [root];
    while (queue.length) {
        const levelSize = queue.length;
        const currentLevel = [];
        for (let i = 0; i < levelSize; i++) {
            const node = queue.shift();
            currentLevel.push(node.val);
            if (node.left) queue.push(node.left);
            if (node.right) queue.push(node.right);
        }
        result.push(currentLevel);
    }
    return result;
}
```

PHP

```php
function levelOrder($root): array {
    if ($root === null) return [];
```

```
    $result = [];
    $queue = [$root];
    while (!empty($queue)) {
        $levelSize = count($queue);
        $currentLevel = [];
        for ($i = 0; $i < $levelSize; $i++) {
            $node = array_shift($queue);
            $currentLevel[] = $node->val;
            if ($node->left !== null) $queue[] = $node->left;
            if ($node->right !== null) $queue[] = $node->right;
        }
        $result[] = $currentLevel;
    }
    return $result;
}
```

Python

```
from collections import deque
def level_order(root):
    if not root:
        return []
    result = []
    queue = deque([root])
    while queue:
        level_size = len(queue)
        current_level = []
        for _ in range(level_size):
            node = queue.popleft()
            current_level.append(node.val)
            if node.left:
                queue.append(node.left)
            if node.right:
                queue.append(node.right)
        result.append(current_level)
    return result
```

Go

```
func levelOrder(root *TreeNode) [][]int {
    result := [][]int{}
    if root == nil {
        return result
    }
    queue := []*TreeNode{root}
    for len(queue) > 0 {
        levelSize := len(queue)
        currentLevel := []int{}
        for i := 0; i < levelSize; i++ {
            node := queue[0]
            queue = queue[1:]
            currentLevel = append(currentLevel, node.Val)
            if node.Left != nil {
                queue = append(queue, node.Left)
```

```
            }
            if node.Right != nil {
                queue = append(queue, node.Right)
            }
        }
        result = append(result, currentLevel)
    }
    return result
}
```

C++

```
vector<vector<int>> levelOrder(TreeNode* root) {
    vector<vector<int>> result;
    if (!root) return result;
    queue<TreeNode*> q;
    q.push(root);
    while (!q.empty()) {
        int levelSize = q.size();
        vector<int> currentLevel;
        for (int i = 0; i < levelSize; i++) {
            TreeNode* node = q.front(); q.pop();
            currentLevel.push_back(node->val);
            if (node->left) q.push(node->left);
            if (node->right) q.push(node->right);
        }
        result.push_back(currentLevel);
    }
    return result;
}
```

SECTION 18 — Dry Runs

Small Input

```
        3
       / \
      9  20
         / \
        15  7
```

```
queue=[3]
Level 0: levelSize=1 → dequeue 3, currentLevel=[3] → enqueue 9,20 → queue=[9,20]
         result=[[3]]
Level 1: levelSize=2 → dequeue 9, currentLevel=[9] → no children
                     → dequeue 20, currentLevel=[9,20] → enqueue 15,7 → queue=[15,7]
         result=[[3],[9,20]]
Level 2: levelSize=2 → dequeue 15, currentLevel=[15] → no children
                     → dequeue 7, currentLevel=[15,7] → no children
         result=[[3],[9,20],[15,7]]
queue empty → done
```

Large Input (Conceptual)

For a complete binary tree with 10^6 nodes, the last level alone can hold ~500,000 nodes — the queue's peak size approaches O(n/2), illustrating the O(w) space cost explicitly, versus a DFS approach on the same tree needing only O(log(10^6)) ≈ 20 stack frames.

Corner Case

`root = null`: return `[]` immediately. Single node `[5]`: `queue=[5]` → Level 0: dequeue 5, currentLevel=[5], no children → result=`[[5]]`, correctly one level containing just the root.

SECTION 19 — Advanced Concepts

- **Coordinate-augmented BFS (Vertical Order Traversal):** enqueue `(node, row, column)` tuples instead of just nodes, incrementing/decrementing column for left/right children — enables grouping by horizontal position while still processing in level order (needed to correctly break ties between nodes at the same column but different rows).
- **BFS for shortest-path advantage:** in "Minimum Depth of Binary Tree," BFS can terminate the instant the first leaf is found, which is provably the shallowest leaf — DFS would need to explore the entire tree (or at least one full root-to-leaf path per branch) to guarantee it found the minimum, making BFS strictly better for this specific "shortest" framing.
- **Overflow-safe width calculation (Maximum Width of Binary Tree):** using position indices (`2*i`, `2*i+1` for left/right children) to compute width can overflow for very deep trees; normalize indices relative to the leftmost index at each level to avoid this.
- **BFS combined with parent-pointer construction:** for "All Nodes Distance K in Binary Tree," first DFS to build a parent-pointer map, then BFS from the target node treating the tree as an undirected graph (allowing "up" traversal via parent pointers) — a powerful cross-pattern combination.

SECTION 20 — Senior Engineer Insights

Staff engineers recognize that the choice between BFS and DFS for tree problems is really a choice about **which structural dimension (depth vs. breadth) the problem's information naturally flows along** — and that this same framing generalizes directly to production systems: monitoring dashboards that aggregate "by time window" (level/breadth-like) versus "by service dependency chain" (depth-like) face an analogous structural choice. Interviewers evaluate whether a candidate can articulate the O(w) vs O(h) space trade-off precisely (not just "BFS uses a queue, DFS uses recursion") and whether they recognize BFS's specific early-termination advantage for shortest-path-style questions — a subtle but important efficiency insight.

SECTION 21 — Cheat Sheet

```
PATTERN: Tree BFS (Level Order Traversal)
RECOGNIZE: "level order," "per level," "level by level," "minimum depth," "side view," "connect next poin
TEMPLATE:
    queue = [root] (if root is not null)
    while queue is not empty:
        levelSize = queue.length         # SNAPSHOT before inner loop
        currentLevel = []
        for i in range(levelSize):
            node = queue.dequeue()
```

```
            currentLevel.append(node.value)
            enqueue non-null children
        result.append(currentLevel)
COMPLEXITY: O(n) time, O(w) space (w = max width)
KEY PROOF: FIFO queue guarantees non-decreasing depth order of dequeuing - levels never interleave
WATCH FOR: queue-size snapshot timing, null-child guards, width-vs-height space trade-off vs DFS
DOESN'T APPLY WHEN: need bottom-up aggregation or path tracking (use DFS instead)
```

SECTION 22 — Revision Notes (5-Minute Review)

- BFS = queue-based, level-by-level; snapshot queue size before each level's inner loop.
- FIFO property guarantees non-decreasing depth order — this is the entire correctness proof.
- Space is O(w) (max width), which can approach O(n) for wide trees — worse than DFS's O(h) for balanced trees.
- Early-termination advantage for shortest-path/minimum-depth problems — BFS finds the shallowest answer first.
- Use DFS instead for bottom-up aggregation (height/sum/diameter) or path tracking.
- Coordinate-augmented BFS (row, column tracking) handles vertical-order/positional variants.

SECTION 23 — Practice Roadmap

Stage	Focus	Recommended LeetCode Problems
Beginner	Basic level order mechanics	Binary Tree Level Order Traversal (102), Average of Levels in Binary Tree (637)
Intermediate	Directional variants, side views	Binary Tree Zigzag Level Order Traversal (103), Binary Tree Right Side View (199), Minimum Depth of Binary Tree (111)
Advanced	Pointer linking, N-ary generalization	Populating Next Right Pointers in Each Node (116/117), N-ary Tree Level Order Traversal (429)
Expert	Coordinate tracking, cross-pattern combination	Vertical Order Traversal of a Binary Tree (987), All Nodes Distance K in Binary Tree (863), Maximum Width of Binary Tree (662)

SECTION 24 — Memory Tricks

- **Mnemonic:** "**S**napshot, **P**rocess, **E**nqueue" (SPE) — Snapshot the level size, Process exactly that many, Enqueue their children.
- **Visualization:** A **rumor spreading hop by hop through a social network** — everyone at the current hop hears it before anyone at the next hop.

- **Recognition shortcut:** “Level,” “layer,” “per row,” “minimum depth” → BFS with a queue, snapshot the level size first.

SECTION 25 — Final Summary

Tree BFS uses a FIFO queue to guarantee nodes are processed in strict non-decreasing depth order, naturally producing level-by-level groupings and offering an early-termination advantage for shortest-path/minimum-depth problems. The single most important thing to remember forever: **always snapshot the queue’s size before the inner processing loop — this is the one technique that correctly separates “this level” from “the next level” being enqueued simultaneously — and always weigh BFS’s O(w) width-dependent space against DFS’s O(h) height-dependent space before defaulting to either.**